

PRACTICAL – 1

Program 1:

```
#include <stdio.h>
int main() {
    char command[100];
    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);
    printf("You entered: %s", command);
    return 0;
}
```

Output:

```
nishanth@LAPTOP-6Q3ON07H:~/practical1$ ./program1
Enter a Linux command: ls
You entered: ls
```

Program 2:

```
#include <stdio.h>
#include <unistd.h>
int main() {
    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
    }
    else if(pid > 0) {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
    }
    else {
        printf("Fork Failed!\n");
    }

    return 0;
}
```

Output:

```
nishanth@LAPTOP-6Q3ON07H:~/practical1$ ./program2
Parent Process
Parent PID : 655
Child Process
Child PID : 656
```

Program 3:

```
#include <stdio.h>
#include <unistd.h>
int main() {
    printf("Executing 'ls -l' command...\n");
    execl("/bin/ls", "ls", "-l", NULL);
    printf("This line will execute only if exec fails.\n");
    return 0;
}
```

Output:

```
nishanth@LAPTOP-6Q3ON07H:~/practical1$ ./program3
Executing 'ls -l' command...
total 60
-rwxr-xr-x 1 nishanth nishanth 16096 Aug  5 03:24 program1
-rw-r--r-- 1 nishanth nishanth  194 Aug  5 03:24 program1.c
-rwxr-xr-x 1 nishanth nishanth 16080 Aug  5 03:30 program2
-rw-r--r-- 1 nishanth nishanth  311 Aug  5 03:30 program2.c
-rwxr-xr-x 1 nishanth nishanth 16000 Aug  5 03:32 program3
-rw-r--r-- 1 nishanth nishanth  214 Aug  5 03:31 program3.c
```

Program 4:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
    pid_t pid = fork();

    if(pid == 0) {
        printf("Child Process Running...\n");
        sleep(3);
        printf("Child Process Finished.\n");
    }
    else {
        printf("Parent Waiting...\n");
        wait(NULL);
        printf("Parent Resumed After Child Completion.\n");
    }

    return 0;
}
```

Output:

```
nishanth@LAPTOP-6Q3ON07H:~/practical1$ ./program4
Parent Waiting...
Child Process Running...
Child Process Finished.
Parent Resumed After Child Completion.
```

Program 5:

```

#include <stdio.h>
#include <unistd.h>
int main() {
pid_t pid = fork();

if(pid == 0) {
    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID : %d\n", getppid());
}
else if(pid > 0) {
    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);
}

return 0;
}

```

Output:

```

nishanth@LAPTOP-6Q3ON07H:~/practical1$ ./program5
Parent Process
Parent PID : 764
Child PID : 765
Child Process
Child PID : 765
Parent PID : 764

```

COMMANDS :

```

shanth@LAPTOP-Q330M07H:~/practical1$ uname -a
Linux LAPTOP-Q330M07H 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
shanth@LAPTOP-Q330M07H:~/practical1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 5 5500U with Radeon Graphics
CPU family:             23
Model:                  184
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               1
BogoMIPS:               4192.11
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm rep_good nopl cpuid extd_apicid tsc
                        known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoex
                        ssbd ibrs ibpb stibp vmcall fsgsbase bmi1 avx2 smep bmi2 rdtseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 clzero xsaveprtr virt_ssbd arat umip rdpid
Virtualization features:
Hypervisor vendor:     Microsoft
Virtualization type:   full
Caches (sum of all):
  L1d:                  192 KiB (6 instances)
  L1i:                  192 KiB (6 instances)
  L2:                   3 MiB (6 instances)
  L3:                   4 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):     0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Old microcode:         Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; untrained return thunk; SMT enabled with STIBP protection
Spec rstack overflow:   Vulnerable; Safe RET, no microcode
Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:            Mitigation; Retpolines; IBPB conditional; STIBP always-on; RSB filling; PBRSSB-eIBRS Not affected; BHI Not affected
Srbds:                 Not affected
Taa:                   Not affected

```

```
nishanth@LAPTOP-6Q3ON07H:~/practical1$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16    0 159.4M  1 disk
sdc   8:32    0    1G    0 disk [SWAP]
sdd   8:48    0    1T    0 disk /mnt/wslg/distro
/

nishanth@LAPTOP-6Q3ON07H:~/practical1$ ps
PID TTY          TIME CMD
 333 pts/0        00:00:00 bash
 808 pts/0        00:00:00 ps

nishanth@LAPTOP-6Q3ON07H:~/practical1$ top
top - 03:40:03 up 20 min,  1 user,  load average: 0.16, 0.05, 0.01
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.8 us,  0.0 sy,  0.0 ni, 98.5 id,  0.8 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 2836.6 total, 2217.3 free,  478.4 used,  216.2 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used. 2358.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24380	15328	11516	S	0.0	0.5	0:01.85	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.1	0:00.05	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.1	0:00.00	init
53	root	19	-1	50396	16784	15484	S	0.0	0.6	0:00.50	systemd-journal
94	systemd+	20	0	22540	14468	11848	S	0.0	0.5	0:00.20	systemd-resolve
96	root	20	0	35116	12160	9164	S	0.0	0.4	0:00.61	systemd-udev
145	root	20	0	2888	1952	1820	S	0.0	0.1	0:00.12	chronyd-starter
146	root	20	0	4472	3060	2812	S	0.0	0.1	0:00.02	cron
147	message+	20	0	8792	5252	4572	S	0.0	0.2	0:00.13	dbus-daemon
153	root	20	0	44092	29472	14332	S	0.0	1.0	0:00.45	networkd-dispat
161	root	20	0	18440	9364	8116	S	0.0	0.3	0:00.21	systemd-logind
184	syslog	20	0	220548	5880	4592	S	0.0	0.2	0:00.15	rsyslogd
220	_chrony	20	0	24444	11924	7168	S	0.0	0.4	0:00.14	chronyd
234	_chrony	20	0	12092	2428	1784	S	0.0	0.1	0:00.01	chronyd
255	root	20	0	5492	2740	2556	S	0.0	0.1	0:00.02	agetty
256	root	20	0	123456	32540	17932	S	0.0	1.1	0:00.44	unattended-upgr
329	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
330	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.14	Relay(333)
333	nishanth	20	0	6268	5600	3928	S	0.0	0.2	0:00.31	bash
334	root	20	0	8648	5504	4660	S	0.0	0.2	0:00.03	login
384	nishanth	20	0	22752	13240	10788	S	0.0	0.5	0:00.32	systemd
386	nishanth	20	0	22912	4112	2108	S	0.0	0.1	0:00.00	(sd-pam)
414	nishanth	20	0	6136	5476	3928	S	0.0	0.2	0:00.06	bash
818	nishanth	20	0	8192	5580	3468	R	0.0	0.2	0:00.00	top

History:

nishanth@LAPTOP-6Q30N07H:~/practical1\$ history

```
1  sudo apt update
2  sudo apt install build-essential
3  gcc --version
4  mkdir OSLab
5  cd OSLab
6  gedit hello.c
7  sudo apt install gedit
8  gedit hello.c
9  sudo apt update
10 sudo apt install gcc
11 gcc --version
12 ls
13 gedit hello.c
14 mkdir OSLab
15 cd OSLab
16 gedit hello.c
17 nano process.c
18 ls
19 gcc process.c -o process
20 ./process
21 uname
22 lscpu
23 lsblk
24 ps
25 top
26 echo $shell
27 echo $SHELL
28 nano demo.c
29 gcc demo.c -o demo
30 ./demo
31 help
32 uname -a
33 mkdir fork
34 cd fork
35 nano program1.c
36 gcc program1.c -o program1
37 nano program1.c
38 gcc program1.c -o program1
39 ./program1
40 nano program2
41 gcc program1.c -o program2
42 ./program2
43 ./program1
44 nano program1.c
45 ls
46 rm program2
47 ls
48 nano program2.c
49 gcc program2.c -o program2
```

```
50 ./program2
51 ls
52 nano program3.c
53 gcc program3.c -o program3
54 ./program3
55 nano program4.c
56 gcc program4.c -o program4
57 ./program4
58 nano program5.c
59 gcc program5.c -o program5
60 ./program5
61 nano program6.c
62 gcc program6 -o program6
63 gcc program6.c -o program6
64 ./program6
65 ./program4
66 nano program7.c
67 gcc program7.c -o program7
68 ./program7
69 nano program8.c
70 gcc program8.c -o program8
71 ./program8
72 cd ~/fork
73 ls
74 cat program1.c
75 ./program1
76 cat program2.c
77 ./program2
78 cat progra3.c
79 cat program3.c
80 ./program3
81 cat program4.c
82 ./program4
83 cat program5.c
84 ./program5
85 cat program6.c
86 ./program6
87 cat program7.c
88 ./program7
89 cat program8.c
90 ./program8
91 history
92 history | sed -n '33,71p'
93 nano program9.c
94 gcc program9.c -o program9
95 ./program9
96 history
97 cd ~/fork
98 ls
99 cat program1.c
```



```
112 ./program1
113 nano program2.c
114 gcc program2.c -o program2
115 ./program2
116 nano program3.c
117 gcc program3.c -o program3
118 ./program3
119 nano program4.c
120 gcc program4.c -o program4
121 ./program4
122 nano program5.c
123 gcc program5.c -o program5
124 ./program5
125 cat program1.c
126 ./program1
127 cat program2.c
128 ./program2
129 cat program3.c
130 ./program3
131 cat program4.c
132 ./program4
133 cat program5.c
134 ./program5
135 mkdir practical1
136 cd practical1
137 nano program1.c
138 gcc program1.c -o program1
139 ./program1.c
140 ./program1
141 cat program1.c
142 nano program2.c
143 gcc program2.c -o program2
144 ./program2
145 nano program3.c
146 gcc program3.c -o program3
147 ./program3
148 nano program4.c
149 gcc program4.c -o program4
150 ./program4
151 nano program5.c
152 gcc program5.c -o program5
153 ./program5
154 uname -a
155 lscpu
156 lsblk
157 ps
158 top
159 ls
160 history
```